```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

```
from google.colab import files
import pandas as pd
uploaded=files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving housing_clean.csv to housing_clean.csv
```

```
df=pd.read_csv('housing_clean.csv')
df.head()
```

	Area	Rooms	Age	Location_Score	Price
0	778	1	6	3	5081030
1	2148	4	27	8	11863210
2	1909	3	3	5	11371453
3	1478	3	16	3	9079771
4	1466	3	14	4	7502204

```
X_simple=df[['Area']]
y=df['Price']
X_tr,X_te,y_tr,y_te=train_test_split(X_simple,y,test_size=0.2,random_state=42)
model_simple=LinearRegression()
model_simple.fit(X_tr,y_tr)
pred_simple=model_simple.predict(X_te)
print('SIMPLE REGRESSION')
print('R2 Score:',round(r2_score(y_te,pred_simple),3))
print('RMSE:',round(np.sqrt(mean_squared_error(y_te,pred_simple)),2))
```

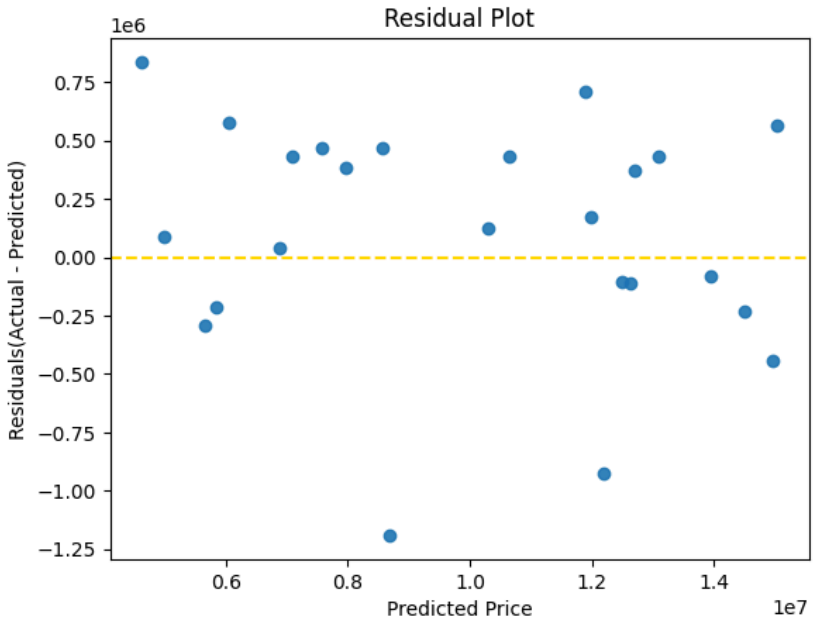
```
SIMPLE REGRESSION
R2 Score: 0.939
RMSE: 809207.7
```

```
X_multi=df[['Area','Rooms','Age','Location_Score']]
X_tr2,X_te2,y_tr2,y_te2=train_test_split(X_multi,y,test_size=0.2,random_state=42)
model_multi=LinearRegression()
model_multi.fit(X_tr2,y_tr2)
pred_multi=model_multi.predict(X_te2)
print('MULTIPLE REGRESSION')
print('R2 Score:',round(r2_score(y_te2,pred_multi),3))
```

```
print('RMSE:',round(np.sqrt(mean_squared_error(y_te2,pred_multi)),2))
```

```
MULTIPLE REGRESSION
R2 Score: 0.977
RMSE: 495063.19
```

```
residual=y_te2-pred_multi
plt.scatter(pred_multi,residual,alpha=0.9)
plt.axhline(y=0,color='GOLD',linestyle='--')
plt.xlabel('Predicted Price')
plt.ylabel('Residuals(Actual - Predicted)')
plt.title('Residual Plot')
plt.show()
```



Start coding or generate with AI.